

Guardrails in Microsoft Foundry





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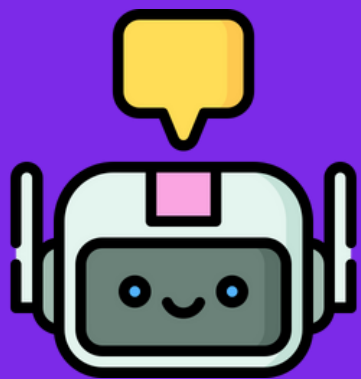


Guardrails = “What is the AI allowed to do, say, or generate?”

Why Guardrails Matter in Production?

Attacks on Agents:

- Prompt injection
- Jailbreak attempts
- Abusive or unsafe user input
- Requests for protected or disallowed content



Microsoft treats guardrails as a first-class, system-level feature in Foundry.

In Foundry, guardrails are implemented through content filtering and risk controls, powered by Azure AI Content Safety.

They work by scanning both:

- user input (prompts)
- model output (responses)

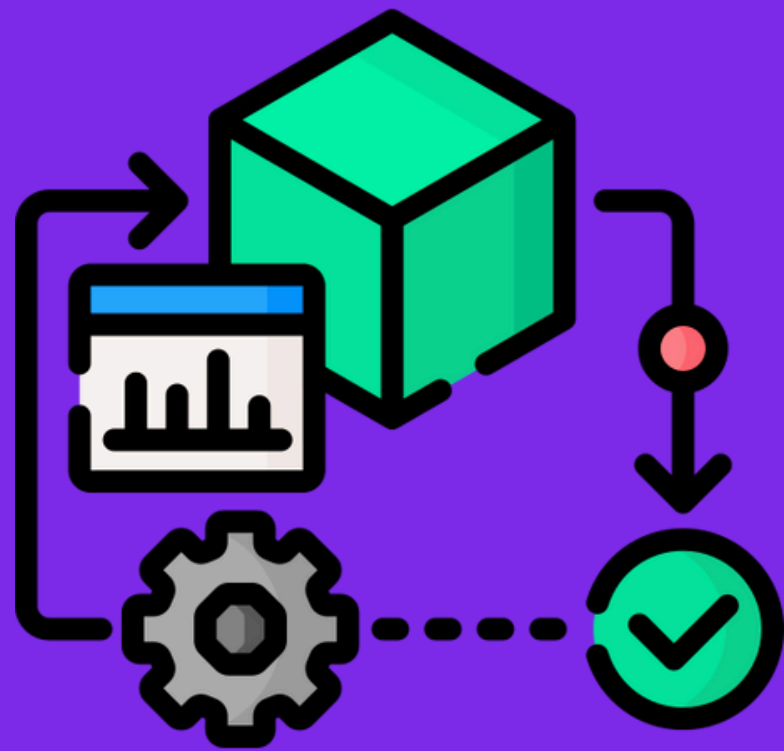
Best Practices

1. Don't weaken the defaults unless you have a very clear reason.
2. Treat guardrails as a global policy layer.
3. Use prompts for behavior. Use guardrails for safety.

Common Pitfalls

1. Trying to “prompt your way” around safety.
2. Disabling guardrails for convenience during demos.
3. Assuming grounding alone prevents misuse.

Mental Model



Prompts define behavior

How the agent thinks, reasons, formats answers, and decides what to do next.

Knowledge defines what it can look up

Approved documents, databases, and sources the agent is allowed to query at runtime.

Grounding defines truth

Ensures responses are based strictly on retrieved knowledge, not assumptions or guesses.

Memory defines what it remembers

Persistent or session-based context from past interactions, used to personalize and stay consistent.

Guardrails define boundaries.

Safety, compliance, and policy constraints that limit what the agent may say or do.

